

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2018

Course: B. Tech. in Civil Engineering

Subject Name: Mechanics of Solids

Date: 03/12/2018

Max Marks: 60

Sem: III

Subject Code: BTCVC302

Duration: 3 Hrs.

*Instructions to the Students:*

1. Solve ANY FIVE questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

COs Marks

**Q. 1 Solve Any Two of the following:**

- A) Define the following properties of materials: Elasticity, Ductility, and Brittleness. CO1 (6)
- B) A metal wire of diameter 3 mm is subjected to an axial tensile force of 2 kN. The extension measured was 4 mm over a length of 1500 mm. Find the modulus of elasticity of the metal. Using the calculated value of modulus of elasticity; find the maximum axial tensile force that can be applied on the wire if the strain is limited to 0.001. CO1 (6)
- C) A thin tyre made up of mild steel is to be shrunk on to a rigid wheel of 1200 mm diameter. Calculate (i) internal diameter of tyre if the hoop stress is limited to  $50 \text{ N/mm}^2$ , and (ii) the least temperature to which the tyre must be heated above that of the wheel before it could be slipped on. For the tyre the coefficient of thermal expansion ( $\alpha$ ) is  $12 \times 10^{-6} \text{ per } ^\circ\text{C}$  and  $E = 2 \times 10^5 \text{ N/mm}^2$ . CO1 (6)

**Q.2 Solve the following.**

- A) Write the assumptions made in the theory of pure bending. CO2 (4)

OR

- A) Find the diameter of a solid shaft which will transmit 150 kW power at 200 r.p.m. if the permissible shear stress is  $60 \text{ N/mm}^2$ . Find also the length of shaft, if the permissible angle of twist is  $1^\circ$  over the entire length. Take, shear modulus =  $80 \times 10^3 \text{ N/mm}^2$  CO2, CO4 (4)
- B) A simply supported beam AB is 10 m long. It carries a uniformly distributed load of 20 kN/m over a distance of 5m from the left end A, a clockwise moment of 50 kN-m at 5m and a point load of 40 kN at a distance of 8 m from the left end A. Find shear force & bending moment at important locations and draw S.F.D. and B.M.D for the beam. CO2 (8)

**Q. 3 Solve the following.**

- A) A masonry pillar square in section 600 mm x 600 mm is subjected to point load of 1800 kN at an eccentricity of 200 mm along one of the centroidal axis of cross section. Find the stresses at four corners. Also determine the maximum eccentricity, if the permissible tensile stress in masonry is limited to 2 N/mm<sup>2</sup>. CO1, CO4 (6)
- B) Find analytically Principal stresses and Principal planes for an element. The element is subjected to two mutually perpendicular stresses 100 N/mm<sup>2</sup> and 50 N/mm<sup>2</sup> both tensile in X and Y direction, respectively along with a shear stress of 30 N/mm<sup>2</sup> (upwards on a plane of 100 N/mm<sup>2</sup> stress). Find also the maximum shear stress. CO1, CO4 (6)

**Q.4 Solve the following.**

- A) Obtain an expression for maximum slope and deflection for a simply supported beam subjected to a central point load. CO3 (4)
- B) A simply supported beam AB of span 6 m is loaded with three point loads 50 kN, 100 kN, and 50 kN each at 1 m, 3 m, and 5 m respectively from left support. Calculate the deflection under each load. Take,  $E = 2 \times 10^5 \text{ N/mm}^2$  and  $I = 20 \times 10^8 \text{ mm}^4$ . CO3 (8)

**Q. 5 Answer the following.**

- A) Obtain an expression for Euler's critical load for a column hinged at both the ends. CO4 (6)
- B) Using Euler's equation for long columns, determine the critical stresses for a compression member of slenderness ratio 80, 120, 160, and 200. The compression member has following end conditions (i) both ends hinged, and (ii) one end hinged and other end fixed.  $E = 2 \times 10^5 \text{ N/mm}^2$ . CO4 (6)

**Q. 6 Answer the following.**

- A) Explain: The Rankine's failure theory. CO1, CO4 (6)
- B) A circular bar is subjected to a tensile force of 20 kN along with a transverse shear force of 10 kN. Determine the diameter of bar using Maximum Principal Stress, Maximum Principal Strain, and Maximum Shear Stress failure theory. Take: Yield strength = 250 MPa, factor of safety = 2, and Poisson's ratio = 0.3 CO1, CO4 (6)

\*\*\*\*\* End of Paper\*\*\*\*\*

**Q. 3 Solve the following.**

- A) A masonry pillar square in section 600 mm x 600 mm is subjected to point load of 1800 kN at an eccentricity of 200 mm along one of the centroidal axis of cross section. Find the stresses at four corners. Also determine the maximum eccentricity, if the permissible tensile stress in masonry is limited to  $2 \text{ N/mm}^2$ . CO1, CO4 (6)
- B) Find analytically Principal stresses and Principal planes for an element. The element is subjected to two mutually perpendicular stresses  $100 \text{ N/mm}^2$  and  $50 \text{ N/mm}^2$  both tensile in X and Y direction, respectively along with a shear stress of  $30 \text{ N/mm}^2$  (upwards on a plane of  $100 \text{ N/mm}^2$  stress). Find also the maximum shear stress. CO1, CO4 (6)

**Q.4 Solve the following.**

- A) Obtain an expression for maximum slope and deflection for a simply supported beam subjected to a central point load. CO3 (4)
- B) A simply supported beam AB of span 6 m is loaded with three point loads 50 kN, 100 kN, and 50 kN each at 1 m, 3 m, and 5 m respectively from left support. Calculate the deflection under each load. Take,  $E = 2 \times 10^5 \text{ N/mm}^2$  and  $I = 20 \times 10^8 \text{ mm}^4$ . CO3 (8)

**Q. 5 Answer the following.**

- A) Obtain an expression for Euler's critical load for a column hinged at both the ends. CO4 (6)
- B) Using Euler's equation for long columns, determine the critical stresses for a compression member of slenderness ratio 80, 120, 160, and 200. The compression member has following end conditions (i) both ends hinged, and (ii) one end hinged and other end fixed.  $E = 2 \times 10^5 \text{ N/mm}^2$ . CO4 (6)

**Q. 6 Answer the following.**

- A) Explain: The Rankine's failure theory. CO1, CO4 (6)
- B) A circular bar is subjected to a tensile force of 20 kN along with a transverse shear force of 10 kN. Determine the diameter of bar using Maximum Principal Stress, Maximum Principal Strain, and Maximum Shear Stress failure theory. Take: Yield strength = 250 MPa, factor of safety = 2, and Poisson's ratio = 0.3 CO1, CO4 (6)

\*\*\*\*\* End of Paper\*\*\*\*\*

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,**  
**LONERE – RAIGAD – 402 103**  
**End Semester Examination – Summer 2019**

**Branch: B. Tech in Civil Engineering**

**Subject Name: Mechanics of Solids**

**Max. Marks: 60**

**Date: - 29/05/2019**

**Semester: III**

**Subject Code: BTCVC 302**

**Duration: 3 Hrs.**

**Instructions to the Students**

1. Attempt any five questions out of the following.
2. The level questions/ expected answer as per OBE or the course outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Illustrate your answers with neat sketches, diagram etc., wherever necessary.

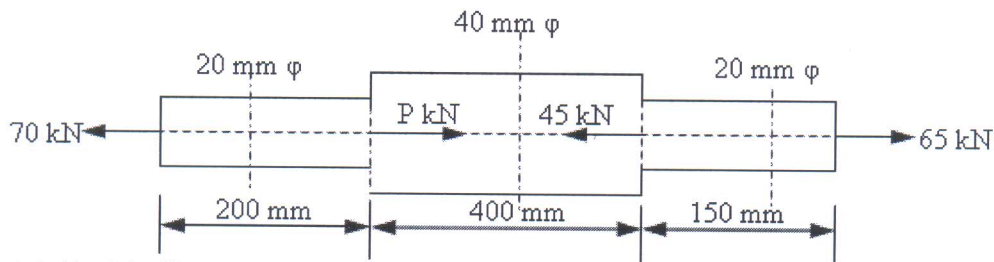
**(Level/ Marks  
CO)**

**Q. 1 (a)** Draw Stress-Strain curve of mild steel. Write characteristic points of the curve.

**L 2      02**

**Q. 1 (b)** A compound composite bar is subjected to forces as shown in the figure 1. Find the force P for equilibrium. Find change in length of the composite bar. The material of 200 mm length is steel, 400 mm length is copper and 150 mm length is alluminium. Use  $E_s = 200GPa$ ,  $E_c = 100GPa$  and  $E_{Al} = 70GPa$

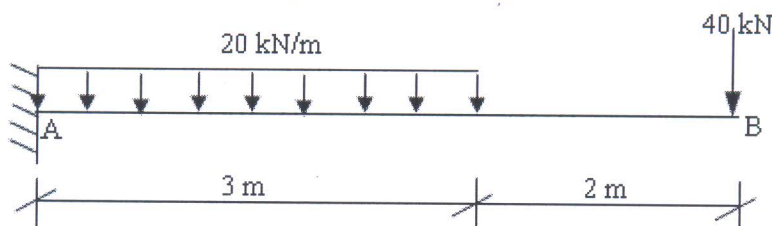
**L 4      10**



**Figure 1**

**Q. 2** Analyse the beam as shown in the figure 2 and hence draw the Shear Force Diagram and Bending Moment Diagram.

**L 4      12**



**Figure 2**

**OR**

- Q. 2 A simply supported beam of span 4 m is subjected to uniformly distributed load of 40 kN/m over entire span. The cross section of beam is T-section which has flange of 200 mm x 40 mm and web of 40 mm x 300 mm. Analyse and draw the maximum flexural stress distribution diagram or shear stress distribution diagram if  $E_s = 210GPa$  L 4 12
- Q. 3 At a certain point, the values of normal stresses across two planes at right angles to each other are  $\sigma_x = 100MPa$  and  $\sigma_y = 40MPa$  both tensile and a shear stress  $\tau = 50MPa$ . Analyse either analytically or Mohr's Circle method and find: L 4 12
1. The Principal Stresses and their locations.
  2. Maximum Shear Stresses and their planes.
- Q. 4 A cantilever beam of span 4 m is subjected to uniformly distributed load of 20 kN/m over length of 2.5 m from free end. Find the slope and deflection at the free end of the beam if  $E = 200GPa$  and  $I = 100 \times 10^6 mm^4$ . Use either Macaulay's method or Conjugate Beam method. L 4 12
- Q. 5 A hollow circular column of height 4 m has 60 mm external diameter and 40 mm internal diameter which is fixed at both ends. Find the ratio of Euler's Load and Rankine's Load carrying capacity of the column if  $E = 200GPa$ ,  $\sigma_c = 320MPa$  and  $a = 1/7500$ . L 4 12
- Q. 6 Write with neat diagram, maximum principal stress theory. Also, write the limitations and applications of this theory. L 4 12

End

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL  
UNIVERSITY, LONERE**

**Winter Semester Examination: Dec.-2019**

**B. Tech. Second Year**

**Mechanics of Solids (BTCVC 302)**

**Date: 12/12/2019**

**Sem.: III**

**Marks: 60**

**Time: 3 Hrs**

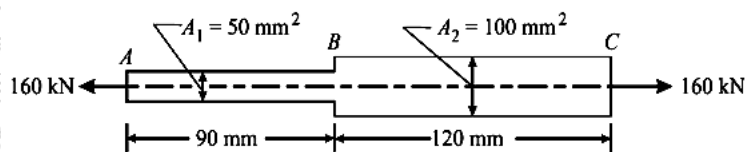
**INSTRUCTIONS TO THE STUDENTS**

1. Attempt any five questions from the given six questions. Figures to the right indicate full marks.
2. Illustrate your answers with neat sketches, diagrams etc. wherever necessary.
3. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that data is a part of the examination.
4. If some part or parameter is noticed to be missing, you may appropriately assume and mention it.
5. Use of logarithmic table, drawing instruments and non programmable calculators is permitted.

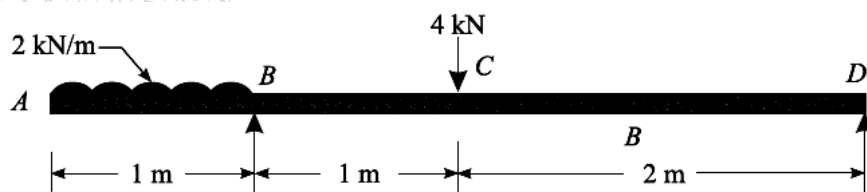
**(Marks)**

**Q. 1** Answer any two from the following:

- a) Define: Hooke's law, Elasticity, and Ductility. (6)
- b) A steel rod 25 mm diameter is at a temperature of  $100^{\circ}\text{C}$ . At this temperature the length of rod is 6 m and is fixed at both of its ends. Determine the stress in the rod, when the temperature falls down to  $60^{\circ}\text{C}$ , if  
(a) the ends do not yield, and (b) the ends yield by 1 mm  
Take  $E = 200 \text{ GPa}$  and  $\alpha = 12 \times 10^{-6}$  per degree Celsius (6)
- c) A steel rod ABC is 210 mm long is subjected to 160 kN force as shown in figure. If the value of Young's modulus for the steel is 200 GPa, determine its deformation. (6)



- Q. 2**
- a) Explain the variation of shear force diagram and bending moment diagram at a point or between any two sections of a beam with respect to type of loading? (4)
  - b) A beam 4 m long is overhanging by 1 m and carries load as shown in figure. Calculate the shear force and bending moment at important locations and draw the SFD and BMD for the beam. Also, locate the point of contraflexure. (8)



- Q. 3**
- a) A solid steel shaft has to transmit 100 kW at 160 r.p.m. Permissible shear stress is 70 MPa. Find the suitable diameter of the shaft. The maximum torque (4)

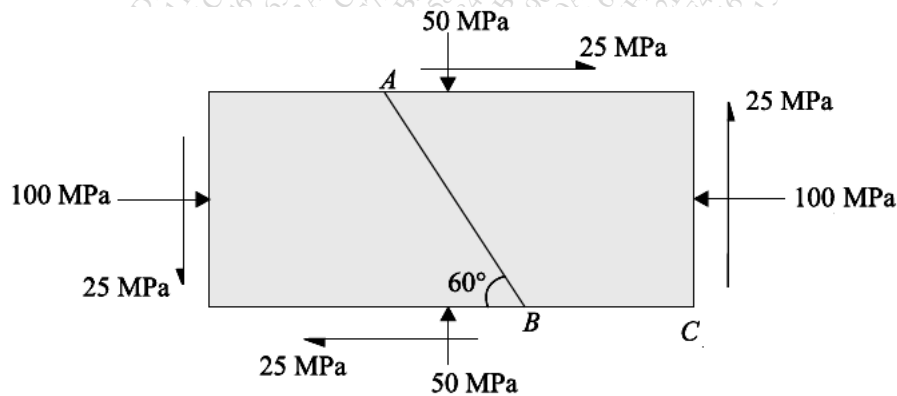
transmitted in each revolution exceeds the mean by 20%.

- b) Write the assumptions made in the theory of simple bending. (4)
- c) A cantilever beam is rectangular in section, width = 80 mm, depth = 120 mm. (4)  
If cantilever is subjected to a point load of 6 kN at free end and the bending stress is not to exceed 40 MPa, find the span of the cantilever beam.

- Q. 4** a) A square section 450 mm x 450 mm is subjected to point load of 900 kN at an eccentricity of 75 mm along one of its axis of cross section. Find the stresses at four corners. Find the maximum uniaxial eccentricity, if permissible tensile stress in masonry is  $2.0 \text{ N/mm}^2$ . (4)

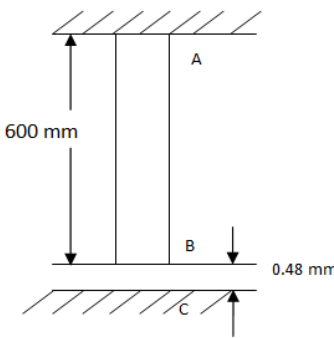
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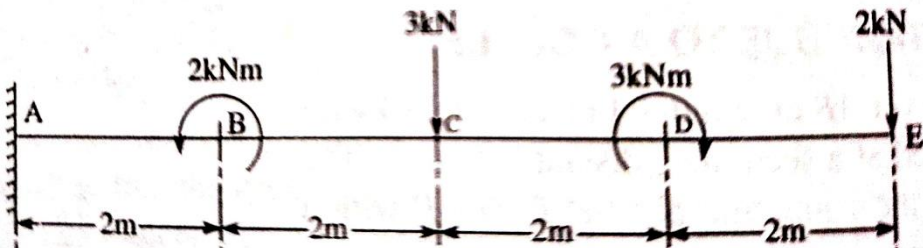
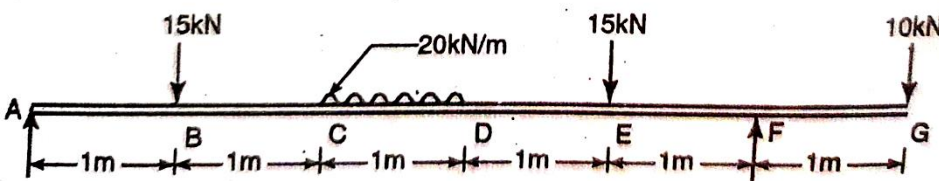
- a) Write the assumptions made in deriving equation of Crippling load using Euler's theory of long columns. (4)
- b) Obtain an expression for Euler's critical load for a column hinged at both ends (8)
- Q. 5** a) A point in a strained material is subjected to the stresses as shown in the figure. Find the normal and shearing stresses on the section AB inclined at an angle of  $60^\circ$  with x-x axis. Also find the resultant stress on the section. (6)



- b) An element in a strained body is subjected to a compressive stress of 200 MPa and a clockwise shear stress of 50 MPa on the same plane. Calculate the values of normal and shear stresses on a plane inclined at  $35^\circ$  with the compressive stress. Also calculate the value of maximum shear stress. (6)
- Q. 6** a) Explain: The maximum Principal Stress theory of failure. (4)
- b) A circular bar is subjected to a tensile force of 20 kN along with a transverse shear force of 10 kN. Determine the diameter of bar using Maximum Principal Stress, Maximum Principal Strain, and Maximum Shear Stress failure theory. Take: Yield strength = 250 MPa, factor of safety = 2, and Poisson's ratio = 0.3 (8)

----- End of Paper -----

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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Winter Examination – 2022</b><br><b>Course: B. Tech.                      Branch : Civil Engineering                      Semester :III</b><br><b>Subject Code &amp; Name: BTCVES302 Mechanics of Solids</b><br><b>Max Marks: 60                      Date:                      Duration: 3 Hr.</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                         |               |           |
| <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |                                                                                                                                                                                                                                                                                                                                                                                                                         |               |           |
|                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                         | (Level/CO)    | Marks     |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                                                        | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                  |               | <b>12</b> |
| A)                                                                                                                                                                                                                                                                                                                                                                                 | Derive the relationship between modulus of elasticity and modulus of rigidity.                                                                                                                                                                                                                                                                                                                                          | <b>L2/CO1</b> | <b>6</b>  |
| B)                                                                                                                                                                                                                                                                                                                                                                                 | A specimen of steel 25mm diameter with a gauge length of 200mm is tested to destruction. It has an extension of 0.16mm under a load of 80 kN and the load at elastic limit is 160 kN. The maximum load is 180 kN. The total extension at fracture is 56mm and diameter at neck is 18mm. Calculate<br>i) The stress at elastic limit<br>ii) Young's Modulus<br>iii) Percentage elongation<br>iv) Ultimate tensile stress | <b>L3/CO1</b> | <b>6</b>  |
| C)                                                                                                                                                                                                                                                                                                                                                                                 | Fig. shows a copper rod AB of length 600mm. When the temperature of the rod is 24 <sup>0</sup> C, the gap of BC is 0.48 mm. Determine<br>i) Thermal stress at 40 <sup>0</sup> C<br>ii) The temperature at which the gap just close<br>iii) Stress and strain in the rod when its temperature is 88 <sup>0</sup> C<br>Take E = 100 GPa and $\alpha = 16 \times 10^{-6}$ per <sup>0</sup> C                               | <b>L3/CO1</b> | <b>6</b>  |
|  <p>The diagram shows a vertical copper rod AB of length 600 mm. The top end A is fixed to a ceiling. The bottom end B is positioned 0.48 mm above a base labeled C. Arrows indicate the dimensions: 600 mm for the rod length and 0.48 mm for the gap between B and C.</p>                     |                                                                                                                                                                                                                                                                                                                                                                                                                         |               |           |

|             |                                                                                                                                                                                                                                                                                                                                                                                            |               |           |
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|             |                                                                                                                                                                                                                                                                                                                                                                                            |               |           |
| <b>Q.2</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                     |               | <b>12</b> |
| A)          | Draw SFD and BMD for a simply supported beam carrying point load at center                                                                                                                                                                                                                                                                                                                 | <b>L3/CO2</b> | <b>6</b>  |
| B)          | Draw Shear force and Bending moment diagram for the beam shown in figure<br>                                                                                                                                                                                                                             | <b>L3/CO2</b> | <b>6</b>  |
| C)          | Draw the SF and BM diagrams for the beam shown in fig. and mark the salient points. Find the point of contra flexure<br>                                                                                                                                                                                | <b>L3/CO2</b> | <b>6</b>  |
|             |                                                                                                                                                                                                                                                                                                                                                                                            |               |           |
| <b>Q. 3</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                     |               | <b>12</b> |
| A)          | Write the assumptions made in the theory of pure bending                                                                                                                                                                                                                                                                                                                                   | <b>L1/CO3</b> | <b>6</b>  |
| B)          | Derive and draw the shear stress distribution diagram for the rectangular cross section of the beam                                                                                                                                                                                                                                                                                        | <b>L2/CO3</b> | <b>6</b>  |
| C)          | A hollow shaft of 60mm outer diameter transmits 180 kW of power while rotating at a frequency of 25 hertz. Find the thickness of the shaft so that the shear stress does not exceed 60 N/mm <sup>2</sup>                                                                                                                                                                                   | <b>L3CO3</b>  | <b>6</b>  |
|             |                                                                                                                                                                                                                                                                                                                                                                                            |               |           |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                     |               | <b>12</b> |
| A)          | Define effective length of column, give its values for various end conditions of column and state the importance of effective length in Euler's formula                                                                                                                                                                                                                                    | <b>L1/CO3</b> | <b>6</b>  |
| B)          | A 1.5m long column has a circular cross section of 50mm diameter, one of the end of the column is fixed in direction and position and the other is free. Taking factor of safety as 3. Calculate the safe load using<br>i) Rankine's Formula with $f_c = 560\text{N/mm}^2$ and $a = 1/1600$ for Pinned end<br>ii) Euler's Formula with $E$ for column = $1.2 \times 10^5 \text{ N/mm}^2$ . | <b>L3CO4</b>  | <b>6</b>  |

|             |                                                                                                                                                                                                                                                                                                                                         |               |           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| C)          | Evaluate the ratio of the buckling strengths of two circular columns one hollow and the other solid. Both the columns are made of the same material and have the same length, cross-sectional area and end conditions. The internal diameter of the hollow column is half of its external diameter.                                     | <b>L3/CO4</b> | <b>6</b>  |
|             |                                                                                                                                                                                                                                                                                                                                         |               |           |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                  |               | <b>12</b> |
| A)          | What is Mohr's stress circle? How is it useful in the solution of stress analysis problems.                                                                                                                                                                                                                                             | <b>L2/CO1</b> | <b>6</b>  |
| B)          | At a point in a piece of elastic material the normal stresses on two mutually perpendicular planes are $80 \text{ N/mm}^2$ tensile and $60 \text{ N/mm}^2$ compressive. These planes also carry shear stresses of $65 \text{ N/mm}^2$ . Determine the principal planes and the principal stresses                                       | <b>L3/CO1</b> | <b>6</b>  |
| C)          | A solid circular shaft is subjected to a bending moment of $40 \text{ kN-m}$ and a torque of $10 \text{ kN-m}$ . Design the diameter of the shaft according to<br>i) Maximum principal stress theory<br>ii) Maximum shear stress theory<br>Take $\mu = 0.25$ , stress at elastic limit = $200 \text{ N/mm}^2$ and factor of safety = 2. | <b>L3/CO1</b> | <b>6</b>  |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                      |               |           |

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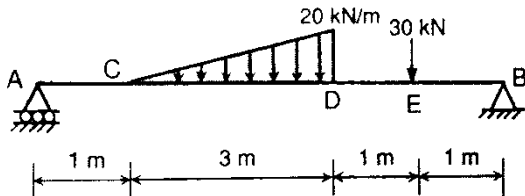
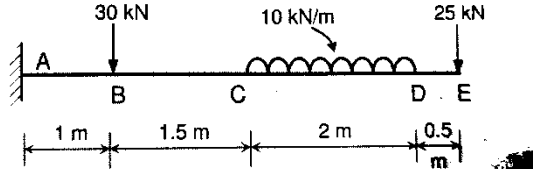
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|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
|             | <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Supplementary Examination – Summer 2022</b><br><b>Course: B. Tech.      Branch :    Civil Engineering      Semester :IV</b><br><b>Subject Code &amp; Name: Mechanics of Solids (BTCVC302)</b><br><b>Max Marks: 60      Date:      Duration: 3 Hr.</b>                                                         |            |           |
|             | <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly. |            |           |
|             |                                                                                                                                                                                                                                                                                                                                                                                    | (Level/CO) | Marks     |
| <b>Q. 1</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                             |            |           |
| <b>A)</b>   | Plot stress strain diagram for mild steel. Explain its salient features.                                                                                                                                                                                                                                                                                                           | <b>CO1</b> | <b>06</b> |
| <b>B)</b>   | A metal wire of diameter 3 mm is subjected to an axial tensile force of 2 kN. The extension measured was 4 mm over a length of 1500 mm. Find the modulus of elasticity of the metal. Using the calculated value of modulus of elasticity; find the maximum axial tensile force that can be applied on the wire if the strain is limited to 0.001.                                  | <b>CO1</b> | <b>06</b> |
| <b>C)</b>   | Define the following terms<br>a) Poissons's ratio    b) Elasticity    c) Hooks law                                                                                                                                                                                                                                                                                                 | <b>CO1</b> | <b>06</b> |
|             |                                                                                                                                                                                                                                                                                                                                                                                    |            |           |
| <b>Q.2</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                             |            |           |
| <b>A)</b>   | A 10 m long simply supported beam carries two point loads of 10kN and 6kN at 2m and 9m respectively from the left end. It also carries a uniformly distributed load of 4kN/m run for the length between 4m to 7m from the left end. Draw shears force and bending moment diagrams.                                                                                                 | <b>CO2</b> | <b>06</b> |
| <b>B)</b>   | A cantilever beam of span L, fixed at the left end, carries a gradually varied load from zero at free end to w per m length at fixed end. Draw the SFD and BMD.                                                                                                                                                                                                                    | <b>CO2</b> | <b>06</b> |
| <b>C)</b>   | Obtain the relationship between bending moment, shear force and load intensity at any section of a beam.                                                                                                                                                                                                                                                                           | <b>CO4</b> | <b>06</b> |
|             |                                                                                                                                                                                                                                                                                                                                                                                    |            |           |
| <b>Q. 3</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                             |            |           |
| <b>A)</b>   | A masonry pillar square in section 600 mm x 600 mm is subjected to point load of 1800 kN at an eccentricity of 200 mm along one of the centroidal axis of cross section. Find the stresses at four corners. Also determine the maximum eccentricity, if the permissible tensile stress in masonry is limited to 2 N/mm <sup>2</sup> .                                              | <b>CO3</b> | <b>06</b> |
| <b>B)</b>   | Find the diameter of a solid shaft which will transmit 150 kW power at 200 r.p.m. if the permissible shear stress is 60 N/mm <sup>2</sup> . Find also the length of shaft, if the permissible angle of twist is 1° over the entire length. Take, shear modulus = 80 x 10 <sup>3</sup> N/mm <sup>2</sup>                                                                            | <b>CO3</b> | <b>06</b> |

|             |                                                                                                                                                                                                                                                                                                                                                                                      |            |           |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
|             |                                                                                                                                                                                                                                                                                                                                                                                      |            |           |
| <b>C)</b>   | A hollow shaft is of external diameter 70 mm and diameter ratio 0.8. It transmits a power of 2 HP at 25 rpm. If the maximum torque exceeds the average torque by 25%, draw the shear stress distribution across the section of the shaft indicating the values.                                                                                                                      | <b>CO3</b> | <b>06</b> |
|             |                                                                                                                                                                                                                                                                                                                                                                                      |            |           |
| <b>Q.4</b>  | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                               |            |           |
| <b>A)</b>   | Derive Rankine's formula for finding the critical load of columns.                                                                                                                                                                                                                                                                                                                   | <b>CO4</b> | <b>06</b> |
| <b>B)</b>   | A circular bar is subjected to a tensile force of 20 kN along with a transverse shear force of 10 kN. Determine the diameter of bar using Maximum Principal Stress, Maximum Principal Strain, and Maximum Shear Stress failure theory. Take: Yield strength = 250MPa, factor of safety = 2, and Poisson's ratio = 0.3                                                                | <b>CO3</b> | <b>06</b> |
| <b>C)</b>   | Using Euler's equation for long columns, determine the critical stresses for a compression member of slenderness ratio 80, 120, 160, and 200. The compression member has following end conditions (i) both ends hinged, and (ii) one end hinged and other end fixed. $E = 2 \times 10^5 \text{ N/mm}^2$ .                                                                            | <b>CO4</b> | <b>06</b> |
|             |                                                                                                                                                                                                                                                                                                                                                                                      |            |           |
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                               |            |           |
| <b>A)</b>   | An elemental cube is subjected to tensile stresses of 30N/mm <sup>2</sup> and 15N/mm <sup>2</sup> acting on two mutually perpendicular planes and a shear stress of 25N/mm <sup>2</sup> on these planes. Determine magnitude and directions of principal stresses.                                                                                                                   | <b>CO4</b> | <b>06</b> |
| <b>B)</b>   | Find analytically Principal stresses and Principal planes for an element. The element is subjected to two mutually perpendicular stresses 100 N/mm <sup>2</sup> and 50 N/mm <sup>2</sup> both tensile in X and Y direction, respectively along with a shear stress of 30 N/mm <sup>2</sup> (upwards on a plane of 100 N/mm <sup>2</sup> stress). Find also the maximum shear stress. | <b>CO4</b> | <b>06</b> |
| <b>C)</b>   | Explain: The Rankine's failure theory.                                                                                                                                                                                                                                                                                                                                               | <b>CO1</b> | <b>06</b> |
|             | <b>*** End ***</b>                                                                                                                                                                                                                                                                                                                                                                   |            |           |

**The grid and the borders of the table will be hidden before final printing.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. Figures to right indicates full marks.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                         | (Level/CO) | Marks     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                             |            | <b>12</b> |
| A) Explain Stress Strain Graph for Mild Steel                                                                                                                                                                                                                                                                                                                                                           | CO-I       | 06        |
| B) Derive the equation of Volumetric Strain for Tri axial loading                                                                                                                                                                                                                                                                                                                                       | CO-I       | 06        |
| C) A steel rod of 20mm diameter and 300mm long is enclosed centrally inside a copper tube of external diameter 30mm and internal diameter 25mm. the ends of tube and rod are brazed (fixed) together. And composite bar is subjected to an axial pull of 40kN. Find the stresses developed in rod and tube . Also find the extension of rod. Take $E_s = 200 \frac{GN}{m^2}$ $E_c = 100 \frac{GN}{m^2}$ | CO-I       | 06        |
| <b>Q.2 Solve Any one of the following.</b>                                                                                                                                                                                                                                                                                                                                                              |            | <b>12</b> |
| A) Draw shear force and Bending moment Diagram for the beam as shown in fig.1, also locate the position and Magnitude of the maximum Bending moment.                                                                                                                                                                                                                                                    | CO-II      | 12        |
|  <p style="text-align: center;"><b>Fig.1</b></p>                                                                                                                                                                                                                                                                     |            |           |
| B) Draw shear force and Bending moment Diagram for the cantilever beam as shown in fig. 2.                                                                                                                                                                                                                                                                                                              | CO-II      | 12        |
|  <p style="text-align: center;"><b>Fig.2</b></p>                                                                                                                                                                                                                                                                    |            |           |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                             |            | <b>12</b> |
| A) Derive Flexural formula                                                                                                                                                                                                                                                                                                                                                                              | CO-III     | 06        |
| B) A cantilever beam of uniform rectangular cross section 150mm x 300mm uniformly distributed load of 20kN/m over its span of 2m. Determine the bending stress anywhere in the beam.                                                                                                                                                                                                                    | CO-III     | 06        |
| C) Find the power that can be transmitted by shaft of 60mm diameter and rotating at 180 r.p.m., If the permissible stress is $85 \frac{N}{mm^2}$ .                                                                                                                                                                                                                                                      | CO-III     | 06        |

|                                                                                                                                                                                                                                                                                                                        |               |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|
| <b>Q.4 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                             |               | <b>12</b> |
| A) Define Strut, column and slenderness ratio                                                                                                                                                                                                                                                                          | <b>CO-III</b> | <b>06</b> |
| B) Calculate the safe compressive load on a hollow C.I. column (one end fixed and other hinged) of 150mm external diameter, 100mm internal diameter and 10m length. Using Euler's formula with factor of safety of 5 and $E=95\text{GPa}$                                                                              | <b>CO-III</b> | <b>06</b> |
| C) The external and internal diameter of a hollow C.I. column are 5cm and 4cm respectively. If the length of its column is 3m and both of its ends are fixed. Determine the crippling load using Rankine's Formula. Take $\sigma_c=550\frac{N}{mm^2}$<br><br>A= 1/1600 in Rankine's Formula.                           | <b>CO-III</b> | <b>06</b> |
| <b>Q. 5 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                            |               | <b>12</b> |
| A) The Tensile Stresses at a point across two mutually perpendicular planes are $100\frac{N}{mm^2}$ and $40\frac{N}{mm^2}$ . Find the graphically the normal and tangential stresses on a oblique plane inclined at $30^\circ$ with the major principal planes. Also find the resultant stress and angle of obliquity. | <b>CO-IV</b>  | <b>06</b> |
| B) A Bar of uniform cross section 60mm x 80mm is subjected to an axial tensile force of 600kN applied at each end of the bar determine the normal and shear stress acting on a plane inclined at $35^\circ$ to the line of action of force.                                                                            | <b>CO-IV</b>  | <b>06</b> |
| C) A point in a strained material is subjected to a tensile stress of $120\frac{N}{mm^2}$ and compressive of $80\frac{N}{mm^2}$ acting at right angles to each other. Find the normal Stress, tangential stress and the resultant on a plane inclined at angle of $30^\circ$ with the compressive stress.              | <b>CO-IV</b>  | <b>06</b> |

**\*\*\* End \*\*\***

**Instructions to the Students:**

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3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (Level/CO)                   | Marks     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------|
| <b>Q. 1 Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                     |                              | <b>12</b> |
| <b>A)</b> A hollow cylinder 2 m long has an outside diameter of 50 mm and inside diameter of 30 mm. If the cylinder is carrying a load of 25 kN, find the stress in the cylinder. Also find the deformation of the cylinder, if the value elastic modulus for the cylinder material is 100 GPa.                                                                                                                                                        | <b>Understand/<br/>CO3</b>   | <b>06</b> |
| <b>B)</b> A steel bar 2 m long and 40 mm in diameter is subjected to an axial pull of 80 kN. Find the length of the 20 mm diameter bore, which should be centrally carried out, so that the total elongation should increase by 20 % under the same pull. Take E for the bar material as 200 GPa.                                                                                                                                                      | <b>Apply/ CO3</b>            | <b>06</b> |
| <b>C)</b> Two parallel walls 6 m apart are stayed together by a steel rod 25 mm diameter passing through metal plates and nuts at each end. The nuts are tightened, when the rod is at a temperature of 100 <sup>0</sup> C. Determine the stress in the rod, when the temperature falls down to 60 <sup>0</sup> C, if<br>a) The ends do not yield, and<br>b) The ends yield by 1 mm<br>Take E = 200 GPa and $\alpha = 12 \times 10^{-6} / ^0 \text{C}$ | <b>Analyze/<br/>CO3</b>      | <b>06</b> |
| <b>Q.2 Solve <u>any two</u> following questions.</b>                                                                                                                                                                                                                                                                                                                                                                                                   |                              | <b>12</b> |
| <b>A)</b> Define and explain the below terms:<br>A.1) Shear Force (SF),<br>A.2) Bending Moment (BM)<br>A.3) Relation between loading, SF, and BM                                                                                                                                                                                                                                                                                                       | <b>Understand<br/>/ CO 2</b> | <b>06</b> |
| <b>B)</b> Draw the <b>shear force and bending moment</b> diagrams for a cantilever beam of span 1.5 m carrying point loads of 2 kN and 1.5 kN at distance of 1 m and 1.5 from fixed end respectively.                                                                                                                                                                                                                                                  | <b>Apply /<br/>CO 2</b>      | <b>06</b> |
| <b>C)</b> A simply supported beam 6 m long is carrying a uniformly distributed load of 5 kN/m over a length of 3 m from the right end. Draw the <b>SF and BM diagrams</b> for the beam and also calculate the <b>maximum BM</b> on the section.                                                                                                                                                                                                        | <b>Analyze /<br/>CO 2</b>    | <b>06</b> |

|             |                                                                                                                                                                                                                                                                                                                                  |                            |           |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| <b>Q. 3</b> | <b>Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                    |                            | <b>12</b> |
| <b>A)</b>   | State and explain the assumptions in the theory of simple bending.                                                                                                                                                                                                                                                               | <b>Understand/<br/>CO2</b> | <b>06</b> |
| <b>B)</b>   | A cantilever beam is rectangular in section having 80 mm width and 120 mm depth. If the cantilever is subjected to a point load of 6 kN at the free end and the bending stress is not to exceed 40 MPa, find the span of the cantilever beam.                                                                                    | <b>Apply / CO2</b>         | <b>06</b> |
| <b>C)</b>   | Calculate the maximum torque that a shaft of 125 mm diameter can transmit, if the maximum angle of twist is 10 in a length of 1.5 m. Take $C = 70$ GPa.                                                                                                                                                                          | <b>Analyze /<br/>CO2</b>   | <b>06</b> |
| <b>Q.4</b>  | <b>Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                    |                            | <b>12</b> |
| <b>A)</b>   | State and explain the assumption in the Euler's column theory.                                                                                                                                                                                                                                                                   | <b>Remember/<br/>CO3</b>   | <b>06</b> |
| <b>B)</b>   | An I section joist 400 mm x 200 mm x 20 mm and 6 m long is used as a strut with both ends fixed. What is Euler's crippling load for the column? Take elastic modulus for the joist as 200 GPa.                                                                                                                                   | <b>Apply /<br/>CO3</b>     | <b>06</b> |
| <b>C)</b>   | Find the <b>Euler's crippling load</b> for a hollow cylindrical steel column of 38 mm external diameter and 2.5 mm thick. Take length of the column as 2.3 and hinged at its both ends. Take $E = 205$ GPa.<br>Also, <b>determine crippling load</b> by Rankine's formula using constants as 1/7500 and yield stress as 335 MPa. | <b>Apply / CO3</b>         | <b>06</b> |
| <b>Q. 5</b> | <b>Solve <u>any two</u> of the following.</b>                                                                                                                                                                                                                                                                                    |                            | <b>12</b> |
| <b>A)</b>   | Define and explain in detail below terms:<br>A.1) Principal planes<br>A.2) Principal stresses                                                                                                                                                                                                                                    | <b>Understand</b>          | <b>06</b> |
| <b>B)</b>   | A wooden bar is subjected to a tensile stress of 5 MPa. What will be the values of normal and shear stresses across a section, which makes an angle of 250 with the direction of the tensile stress.                                                                                                                             | <b>Apply</b>               | <b>06</b> |
| <b>C)</b>   | Enlist the different theories of failure and explain in details one of the theory.                                                                                                                                                                                                                                               | <b>Understand</b>          | <b>06</b> |

**\*\*\* End \*\*\***

Course: B. Tech.

Branch: Civil Engineering/Civil & Environmental Engg

Subject Code & Name: BTCECS302/BTCVES302 & Mechanics of Solids

Semester: III

Max Marks: 60

Date: 04/01/2024

Duration: 3 Hr.

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

**Q. 1 Solve Any Two of the following.**

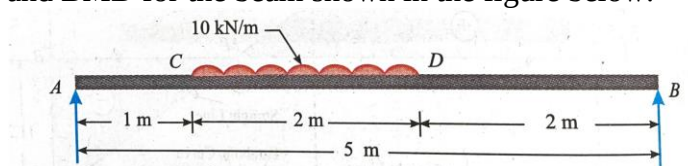
12

- A) Derive the relationship between modulus of elasticity and modulus of rigidity. **L2/CO1** **6**
- B) An Aluminum block 50 mm long placed over a Steel block 150 mm long. **L3/CO1** **6**  
Both blocks have same cross-sectional area  $A = 900 \text{ mm}^2$ . The assembly carries a load  $P$  which deforms it by 0.30 mm. Determine the load  $P$ .  
Take  $E_{\text{aluminium}} = 7 \times 10^4 \text{ N/mm}^2$  and  $E_{\text{steel}} = 2 \times 10^5 \text{ N/mm}^2$ .
- C) A rectangular bar 300 mm long, 80 mm wide and 20 mm thick is loaded with **L3/CO1** **6**  
an axial tensile load of 180 kN; together with a normal compressive force of 1800 kN on  $80 \text{ mm} \times 300 \text{ mm}$  face and tensile force of 300 kN on  $20 \text{ mm} \times 300 \text{ mm}$  face. Calculate change in length, width, thickness and volume.  
Assume  $E = 200 \text{ GPa}$  and  $\mu = 0.3$

**Q.2 Solve Any Two of the following.**

12

- A) Draw the shear force diagram and bending moment diagram for a cantilever **L3/CO2** **6**  
beam carrying a uniformly distributed load from its free end to mid-point.
- B) A horizontal beam AD, 15 m long carries a udl of 1.6 kN/m over whole span **L3/CO2** **6**  
AD together with a concentrated load of 4 kN at the left end A. The beam is supported at a point B which is 1 m from A and at C which is 11 m from left end A. Draw SFD and BMD for this beam.
- C) Draw SFD and BMD for the beam shown in the figure below: **L3/CO2** **6**



**Q. 3 Solve Any Two of the following.**

12

- A) Define the neutral axis in the context of pure bending. Discuss how it relates **L2/CO3** **6**  
to the distribution of bending stress.

- |    |                                                                                                                                                                                                                                          |               |          |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|
| B) | A 2.5 mm thick high strength copper strip 2.5 m long is bend into a circle and held with its ends just in contact. Find the maximum bending stress in the strip. How does this stress change if the thickness of the strip is increased? | <b>L3/CO3</b> | <b>6</b> |
| C) | A hollow shaft is to transmit 400 kW at 160 r.p.m. If the shear stress is not to exceed 120 MPa and internal diameter is 0.6 of the external diameter, find the diameters of the shaft.                                                  | <b>L3/CO3</b> | <b>6</b> |

**Q.4 Solve Any Two of the following.** **12**

- |    |                                                                                                                                                                                                                                                                                                                                                                                                                     |               |          |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|
| A) | Define the terms "slenderness ratio" and "effective length" in the context of columns.                                                                                                                                                                                                                                                                                                                              | <b>L1/CO4</b> | <b>6</b> |
| B) | A short length of tube 36 mm outside diameter and 24 mm inside diameter is tested under axial compression. In this test, tube failed at a load of 175 kN without buckling. When the tube of same section was tested as a strut with $L=1500$ mm both ends hinged, the Euler's load noted was 75 kN. Determine Rankine's constant.                                                                                   | <b>L3/CO4</b> | <b>6</b> |
| C) | A hollow cast iron column whose outer diameter is 200 mm has a thickness of 20 mm. It is 4.5 meters long and is fixed at both the ends. Calculate the safe load by Rankine's formula using a factor of safety of 4. Calculate the slenderness ratio and the ratio of Euler's and Rankine's critical loads. For cast iron take $F_c = 550 \text{ N/mm}^2$ and Rankine's constant = $1/1600$ and $E = 80 \text{ GPa}$ | <b>L3/CO4</b> | <b>6</b> |

**Q. 5 Solve Any Two of the following.** **12**

- |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |               |          |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|
| A) | Explain the maximum principal stress theory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>L2/CO1</b> | <b>6</b> |
| B) | A plane element in a boiler is subjected to tensile stresses of 400 MPa on one plane and 150 MPa on the other at right angles to the former. Each of the above stresses is accompanied by a shear stress of 100 MPa such that when associated with the minor tensile stress tends to rotate the element in anticlockwise direction. Find <div style="margin-left: 20px;">           i. Principal stresses and their directions<br/>           ii. Maximum shearing stresses and the directions of the plane on which they act         </div> | <b>L3/CO1</b> | <b>6</b> |
| C) | The stresses at a point of a machine components are 150 MPa and 50 MPa both tensile. Find the intensities of normal, shear and resultant stresses on a plane inclined at an angle of $55^\circ$ with the axis of major tensile stress. Also find the magnitude of the maximum shear stresses in the components.                                                                                                                                                                                                                              | <b>L3/CO1</b> | <b>6</b> |

**\*\*\* End \*\*\***